PALADIN
BLOCKCHAIN SECURITY

Smart Contract Security Assessment

Final Report

For LayerZero (DVN Differential Audit)

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1 Overview

The DVN contracts at commit `6356f0a` on the public LayerZero-v2 (https://github.com/LayerZero-Labs/LayerZero-v2) repository are derived from the "Verifier Network" contracts audited by Paladin in August 2023. We performed a line-by-line diff between the final audited commit (`f8b2e33` on the private monorepo) and the target commit (`6356f0a`) to assess whether the audited security properties are preserved.

Result: All core security-critical logic is the same. The changes between the two commits fall into four categories:

1. **Rename**: `VerifierNetwork` -> `DVN`; `VerifierFeeLib` -> `DVNFeeLib` and all related identifiers

2. **Error handling:** require() with string messages replaced by custom revert errors

3. **Cosmetic:** `uint` -> `uint256`, import style changes, license/pragma updates

4. **Functional (low risk)**: `verifyAndDeliver()` removed, `nativeDecimalsRate` made constructor-configurable, `withdrawToken()` added to `Worker`, `VerifierFeePaid` event added

At `6356f0a`, `MultiSig` signer storage (`mapping(address => bool)`) and `DVNFeeLib` fee constants (`EXECUTE_FIXED_BYTES = 68, UPDATE_HASH_BYTES = 224`) remain unchanged from the audited version.

On current main, all three DVN-scope contracts have been further modified – `DVN.sol` gained `lzRead`/`CmdLib` overloads, `MultiSig` migrated to `EnumerableSet` storage, and `DVNFeeLib` added `lzRead` support with updated fee constants. These changes are analyzed in Section 2.4.

1.1 Summary

Verification of code equivalence between audited VerifierNetwork and public DVN contracts.

Original Audit	Paladin Blockchain Security (https://github.com/LayerZero-Labs/Audits/blob/main/audits/DVN/DVN-Paladin-26AUG2023.pdf), 26 August 2023
Originally Audited Name	"Verifier Network" (now "DVN")
Target Commit	`6356f0a` (https://github.com/LayerZero-Labs/LayerZero-v2/commit/6356f0a31f2125267420e7bf8403b59a55459aa3)

1.2 Scope

Original Audit Scope: Three contracts in packages/layerzero-v2/evm/messagelib/contracts/ on the private monorepo:

- `VerifierNetwork.sol` (now `DVN.sol`)
- `Worker.sol`
- `MultiSig.sol`

Diff Target: Commit `6356f0a31f2125267420e7bf8403b59a55459aa3` (https://github.com/LayerZero-Labs/LayerZero-v2/commit/6356f0a31f2125267420e7bf8403b59a55459aa3) on the public LayerZero-v2 repository.

1.3 Rename Mapping

At commit `6356f0a` the contracts already carry their public names. The table below maps each audited file to its current counterpart.

Audited Name (f8b2e33)	Name at 6356f0a	Change
`uln/VerifierNetwork.sol`	`uln/dvn/DVN.sol`	Renamed + moved to subdirectory
`uln/VerifierFeeLib.sol`	`uln/dvn/DVNFeeLib.sol`	Renamed + moved
`uln/MultiSig.sol`	`uln/dvn/MultiSig.sol`	Moved to subdirectory
`Worker.sol`	`Worker.sol`	Unchanged
`uln/interfaces/IVerifier.sol`	`uln/interfaces/IDVN.sol`	Renamed
`uln/interfaces/IVerifierFeeLib.sol`	`uln/interfaces/IDVNFeeLib.sol`	Renamed
`uln/interfaces/ILayerZeroVerifier.sol`	`uln/interfaces/ILayerZeroDVN.sol`	Renamed
`uln/interfaces/IUlnConfig.sol`	`*(removed)*`	No longer needed
`uln/interfaces/IUltraLightNode.sol`	`uln/interfaces/IReceiveUlnE2.sol`	Renamed
`uln/libs/VerifierOptions.sol`	`uln/libs/DVNOptions.sol`	Renamed

Additional code-level renames: contract `VerifierNetwork` -> contract `DVN`; `IVerifier` -> `IDVN`; `IVerifierFeeLib` -> `IDVNFeeLib`; license header BUSL-1.1 -> LZBL-1.2, pragma ^0.8.19 -> ^0.8.20.

1.4 Commit Timeline

Commit	Version	Repository	Phase
`bddcada`	v1.5.33	monorepo (private)	Preliminary
`084abac`	v1.5.38	monorepo (private)	Resolution #1
`f8b2e33`	v1.5.38	monorepo (private)	Resolution #2 - Final
`4a522bac`	v1.5.70	monorepo (private)	Post-audit transition
`6356f0a`	**-**	**LayerZero-v2 (public)**	**Target commit**

1.5 Security Fixes During the Original Audit

For completeness, the following security-relevant changes were made during the audit resolution process. All fixes remain intact at 6356f0a.

Preliminary -> Resolution #1:

- Added vid (Verifier Instance ID) to prevent cross-instance replay attacks. Addresses finding #01.
- Replaced raw `ecrecover` with OpenZeppelin `ECDSA.tryRecover` for signature malleability protection. Addresses finding #08.
- Restructured `execute()` to set hash before execution, preventing reentrancy-based re-execution.
- Replaced manual bool paused with OpenZeppelin `Pausable`. Addresses finding #06.
- Added zero-address guard for `roleAdmin` grants. Addresses finding #04.
- `VerifierFeeLib.sol` unchanged across all audit commits.

Resolution #1 -> Resolution #2 (Final):

- Reordered `execute()` steps: signatures are now checked before the hash is set. This prevents gas griefing where invalid submissions could temporarily block valid ones. Related to finding #01.
- Simplified `MultiSig` error handling: three distinct signature error types consolidated into a single `SignatureError`.
- `Worker.sol` and `VerifierFeeLib.sol` unchanged.

1.6 Summary

Contract	f8b2e33 -> 6356f0a	6356f0a -> main	Assessment
`DVN.sol`	Rename + custom errors + verifyAndDeliver removed	lzRead/CmdLib overloads added (349->393 lines)	Low risk - ULN path unchanged
`MultiSig.sol`	Custom errors + moved to subdirectory	EnumerableSet storage rewrite	Low risk - core verification preserved
`Worker.sol`	Rename + custom errors + additive functions	Identical	Low risk
`DVNFeeLib.sol`	Rename + custom errors + nativeDecimalsRate configurable	Fee constants updated + lzRead support added	Low risk - ULN path preserved

All changes between the final audited commit (`f8b2e33`) and the target commit (`6356f0a`) are rename/cosmetic or low-risk functional additions.

Changes between `6356f0a` and main for `DVN.sol`, `MultiSig` and `DVNFeeLib` are documented in Section 6.4. The security properties established by the original Paladin audit are preserved.

2 Diff Analysis

2.1 Core Logic Preservation

The security-critical paths are preserved:

Function	Status
`execute() - sigs -> hash -> exec flow`	Preserved (custom errors only)
`quorumChangeAdmin() - vid check + multisig`	Preserved
`assignJob() / getFee() - fee calculation`	Preserved
`verifySignatures() - ECDSA.tryRecover loop`	Preserved
`_setSigner() - mapping(address => bool) storage`	Preserved (not rewritten to EnumerableSet)

The `execute()` function retains the exact sigs -> hash -> exec ordering from the Resolution #2 fix.

2.2 Cosmetic Changes

The following carry no security impact: contract/interface renames (see Section 1.3), `require()` -> custom revert errors; `uint` -> `uint256`, import style changes (string paths -> { named } imports), license and pragma updates, comment style (`// @notice -> /// @notice`) and event parameter types (`uint idx` -> `uint256 idx`).

The interface selector `IUltraLightNode.verify.selector` was renamed to `IReceiveUlnE2.verify.selector` but retains the same value (`0x0223536e`).

2.3 Functional Changes

- *(Low risk)* `verifyAndDeliver()` removed from DVN.sol.
  This combined function called both verify and deliver in a single transaction. It was a gas optimization; verify and deliver are now called separately. The underlying ULN deliver operation is idempotent.

- `VerifierFeePaid` event added in `DVN.sol`.
  Emitted in the ULNv2 `assignJob()` overload for backward compatibility. Event-only, no logic change.

- `nativeDecimalsRate` made configurable in `DVNFeeLib.sol`. Changed from uint internal constant `NATIVE_DECIMALS_RATE = 1e18` to `uint256` private immutable `nativeDecimalsRate` (set in constructor). This supports chains with non-18-decimal native tokens. Being immutable, it cannot be changed after deployment. The fee calculation constants are unchanged: `EXECUTE_FIXED_BYTES = 68`, `SIGNATURE_RAW_BYTES = 65`, `UPDATE_HASH_BYTES = 224`.

- Parameter types changed in `DVNFeeLib.sol`.
  `memory` -> `calldata` for gas optimization. `_decodeVerifierOptions()` renamed to `_decodeDVNOptions()`. The options decoder now reverts on any unrecognized option type (previously had a todo comment for precrime handling).

- Gas zero check and `receive()` added in `DVNFeeLib.sol`.
  Added `if (_dstConfig.gas == 0) revert DVN_EidNotSupported(...)` as a defensive check in `getFeeOnSend()` and `getFee()`. Added `receive() external payable {}` to accept native token transfers.

- *(Low risk)* `Worker.sol` additions. Added `withdrawToken()` (admin-only, uses `Transfer.nativeOrToken`), `supportedOptionTypes` mapping with getter/setter and extracted `hasAcl()` as a public view function. Import changed from `IMessageLib` to `ISendLib`. All additions use existing `ADMIN_ROLE` access control.

2.4 Changes Between 6356f0a and main

The following documents the diff between the target commit (6356f0a) and current main for DVN.sol, MultiSig.sol and DVNFeeLib.sol. DVN Adapters (CCIPDVNAdapter, AxelarDVNAdapter, DVNAdapterBase) and Read Channel/ CmdLib contracts (ReadLib1002, etc.) are excluded from this analysis.

MultiSig.sol (104 -> 135 lines)

The signer storage was rewritten from mapping(address => bool) public signers + uint64 public signerSize to an EnumerableSet.AddressSet internal signerSet. This is the most significant structural change.

Area	6356f0a	main
`Signer storage`	mapping(address => bool) + uint64 signerSize	EnumerableSet.AddressSet
`Constructor`	Requires sorted signers, reverts on signer <= lastSigner	No ordering requirement, uses signerSet.add(), adds address(0) check
`_setSigner()`	Single function with add/remove logic	Rewritten with separate add/ remove paths via signerSet.add()/ signerSet.remove()
`verifySignatures()`	signers[currentSigner] lookup	isSigner(currentSigner) (functionally equivalent)
`View functions`	signers(address), signerSize()	getSigners(), signers(address) (backward compat), isSigner(address), signerSize()
`Errors`	-	Added MultiSig_StateNotSet, MultiSig_InvalidSigner

The core signature verification logic - the `ECDSA.tryRecover` loop, quorum check and ordered-signer deduplication - is preserved. The `EnumerableSet` provides enumeration capability but changes the storage layout. Removing the sorted-signer constructor requirement is safe since `add()` handles duplicates internally. The `signers(address)` `view` function maintains backward compatibility with the old mapping interface.

DVNFeeLib.sol (145 -> 288 lines)

The contract nearly doubled in size, primarily due to lzRead support. The core ULN fee calculation path is preserved.

Area	6356f0a	main
`EXECUTE_FIXED_BYTES`	68	260
`UPDATE_HASH_BYTES`	224	Removed
`VERIFY_BYTES_ULN`	-	288 (new)
`VERIFY_BYTES_CMD_LIB`	-	288 (new)
`nativeDecimalsRate`	`private immutable`	internal immutable
`Constructor`	(`uint256 _nativeDecimalsRate`)	Added `_localEidV2` parameter
`getFeeOnSend()`	Direct calculation	Delegates to `getFee()` for ULN; new overload for Read operations
`getFee()`	external view	public view; new overload with cmdFee calculation for Read
`_getCallDataSize()`	Direct calculation	Calls `_getCallDataSizeByQuorumAndVerifyBytes()` with VERIFY_BYTES_ULN
`PriceFeed call`	`estimateFeeOnSend`	Renamed to `estimateFeeByEid`

New additions on main: `_getReadCallDataSize()`, `_getCallDataSizeByQuorumAndVerifyBytes()`, `_estimateCmdFee()`, `_assertCmdTypeSupported()` (internals); `setSupportedCmdTypes()`, `setDstBlockTimeConfigs()`, `setCmdFees()` (admin); `getSupportedCmdTypes()`, `getCmdFees()`, `version()` (views); and new state variables `localEidV2`, supportedCmdTypes, `evmCallRequestV1FeeUSD`, `evmCallComputeV1ReduceFeeUSD`, `evmCallComputeV1MapBps`, `dstBlockTimeConfigs`.

The fee constant changes (68->260, 224->288) affect gas cost estimation conservatism, not security logic - higher values mean more conservative gas budgets. The `lzRead` additions are entirely additive and do not modify the existing ULN fee path.

DVN.sol (349 -> 393 lines)

The contract gained `lzRead/CmdLib` support. Changes are entirely additive - no existing functions were modified.

Area	6356f0a	main
`localEidV2`	-	New uint32 public immutable for read call routing
`Constructor`	4 params	Added `_localEidV2` parameter
`assignJob()`	1 overload (ULN)	Added 2nd overload for CmdLib (read operations)
`getFee()`	2 overloads (ULN)	Added 3rd overload for CmdLib (read operations)
`_shouldCheckHash()`	Checks verify.selector	Also checks `ReadLib1002.verify.selector`
`Return style`	return keyword	Named return variable fee (cosmetic)
`import`	-	Added `ReadLib1002` import

(Low risk) The existing ULN execution path (`assignJob`, `getFee`, `execute`, `verifySignatures`) is unchanged. The new `CmdLib` overloads delegate to `IDVNFeeLib.getFeeOnSend/getFee` with `FeeParamsForRead` struct, mirroring the ULN pattern. The `ReadLib1002.verify.selector` addition to `_shouldCheckHash()` extends the replay-safe function list for read operations.

2.5 SendUln302 & ReceiveUln302

During the audit period, the ULN302 contract existed as a monolithic `UltraLightNode302.sol`. It was split into `SendUln302.sol` and `ReceiveUln302.sol` during the post-audit transition (commit `4a522bac`). Both are present at `6356f0a` and identical to current main.

These contracts were not in the scope of the Paladin DVN audit. They are noted here for completeness, as `DVN.sol` interacts with them via the `verify()` call path. Full coverage of `ULN302` and the broader `messagelib` contracts is provided in the MessageLib Diff Audit Report.

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